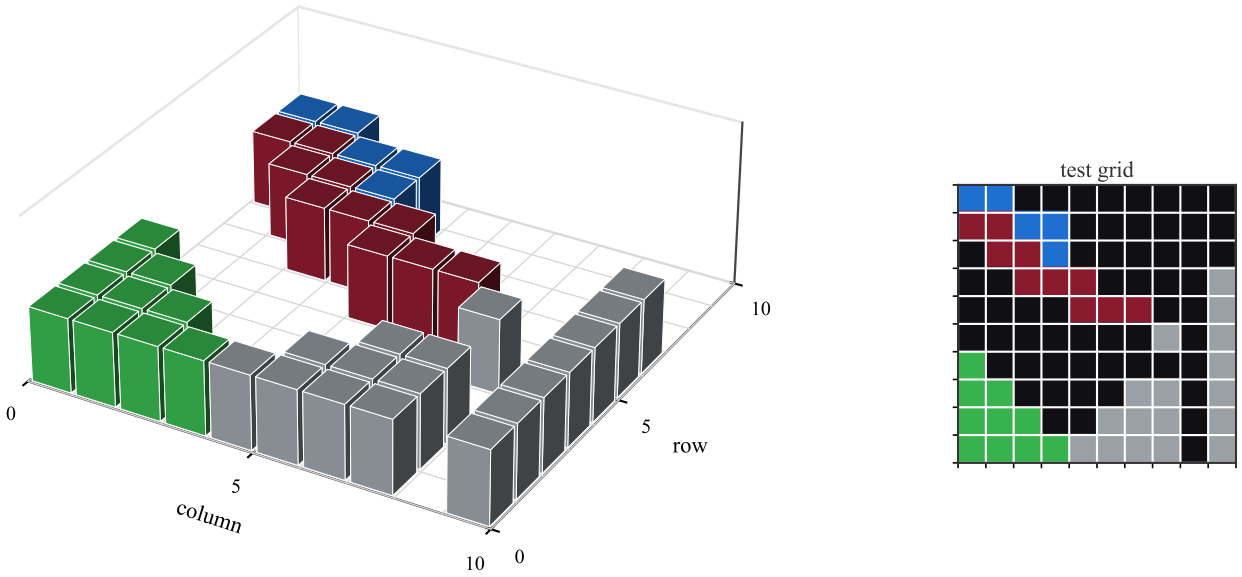
Test-time training on a single consumer GPU

How far the 2025 ARC-AGI-2 winning recipe goes on one 16 GB RTX 5070 Ti



as 3D cells. Background cells omitted.

Summary

We ask how far the 2025 ARC-AGI-2 winning recipe, per-task test-time training (TTT) with augment-inference-reverse-vote (AIRV) decoding, can be pushed on a single 16 GB consumer GPU. The binding constraint turns out to be the serving engine, not the model: moving from eager decoding to llama.cpp raises throughput roughly 13x, from 13 to 170 tokens per second, which is what makes per-task TTT feasible at all. A no-TTT AIRV baseline scores 12.79% on the 120-task public evaluation. Training a small regularized adapter per task and tuning its scale lifts accuracy to 17.6% on the 84 tasks that fit in memory, against 12.8% without TTT, for a best blended result of 16.28% across the full 120. Colour-permutation augmentation lifts the baseline but does not stack with TTT, whose adapter is already colour-robust. The contribution is an honest single-GPU operating point and a map of the walls, the decode ceiling, the VRAM limit that excludes large grids, and adapter overfitting, reported as accuracy per GPU-hour.

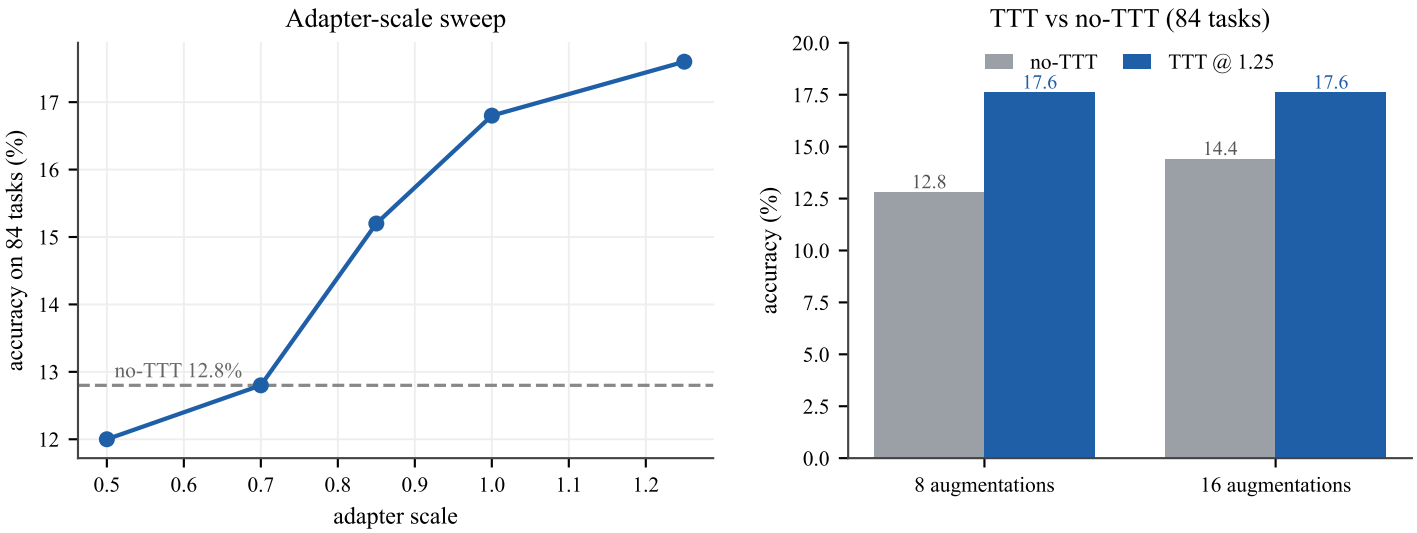


Figure 2. Left: accuracy rises with adapter scale, overtaking the no-TTT baseline. Right: colour-perm lifts the baseline but does not stack with TTT.